In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

In [2]:
```
Forest_fires = pd.read_csv('amazon.csv', encoding='latin1')
```

In [3]:
```
Forest_fires.head()
```

Out[3]:

	year	state	month	number	date
0	1998	Acre	Janeiro	0.0	1998-01-01
1	1999	Acre	Janeiro	0.0	1999-01-01
2	2000	Acre	Janeiro	0.0	2000-01-01
3	2001	Acre	Janeiro	0.0	2001-01-01
4	2002	Acre	Janeiro	0.0	2002-01-01

In [4]:
```
Forest_fires.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6454 entries, 0 to 6453
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   year    6454 non-null   int64
 1   state   6454 non-null   object
 2   month   6454 non-null   object
 3   number  6454 non-null   float64
 4   date    6454 non-null   object
dtypes: float64(1), int64(1), object(3)
memory usage: 252.2+ KB
```

In [5]:
```
Forest_fires.isnull().sum()
#no null values
```

Out[5]:
```
year      0
state     0
month     0
number    0
date      0
dtype: int64
```

In [6]:
```
Forest_fires['number'].describe()
```

Out[6]:
```
count    6454.000000
mean      108.293163
std       190.812242
min         0.000000
25%         3.000000
50%        24.000000
75%       113.000000
max       998.000000
Name: number, dtype: float64
```

In [7]:
```
Forest_fires.year.unique()
```

Out[7]:
```
array([1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008,
       2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017], dtype=int64)
```

In [8]:
```
Forest_fires.month.unique()
```

Out[8]:
```
array(['Janeiro', 'Fevereiro', 'Março', 'Abril', 'Maio', 'Junho', 'Julho',
       'Agosto', 'Setembro', 'Outubro', 'Novembro', 'Dezembro'],
      dtype=object)
```

In [9]:
```
Forest_fires.state.unique()
```

Out[9]:
```
array(['Acre', 'Alagoas', 'Amapa', 'Amazonas', 'Bahia', 'Ceara',
       'Distrito Federal', 'Espirito Santo', 'Goias', 'Maranhao',
       'Mato Grosso', 'Minas Gerais', 'Pará', 'Paraiba', 'Pernambuco',
       'Piau', 'Rio', 'Rondonia', 'Roraima', 'Santa Catarina',
       'Sao Paulo', 'Sergipe', 'Tocantins'], dtype=object)
```

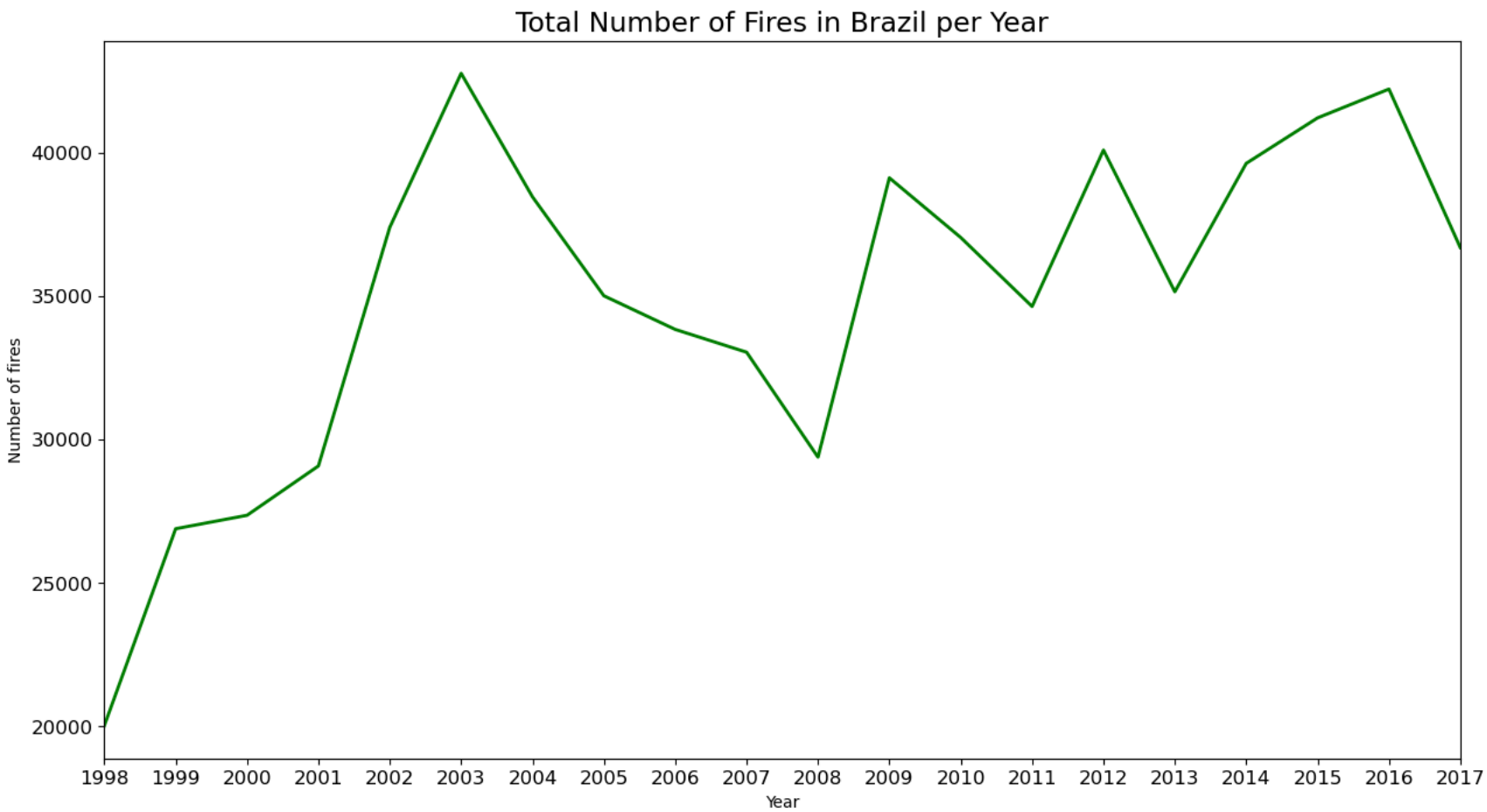
In [10]:
```
ffpy = pd.DataFrame(Forest_fires.groupby('year')['number'].sum().reset_index())

plt.figure(figsize=(15,8))
sns.lineplot(x='year', y='number', data=ffpy, color='green', lw=2)
plt.xlabel('Year')
plt.ylabel('Number of fires')
plt.title('Total Number of Fires in Brazil per Year', fontsize=17)
plt.xlim(1998,2017)
```

```
plt.xticks(np.arange(1998, 2018, 1),fontsize=12)
plt.yticks(fontsize=12)
```

Out[10]:

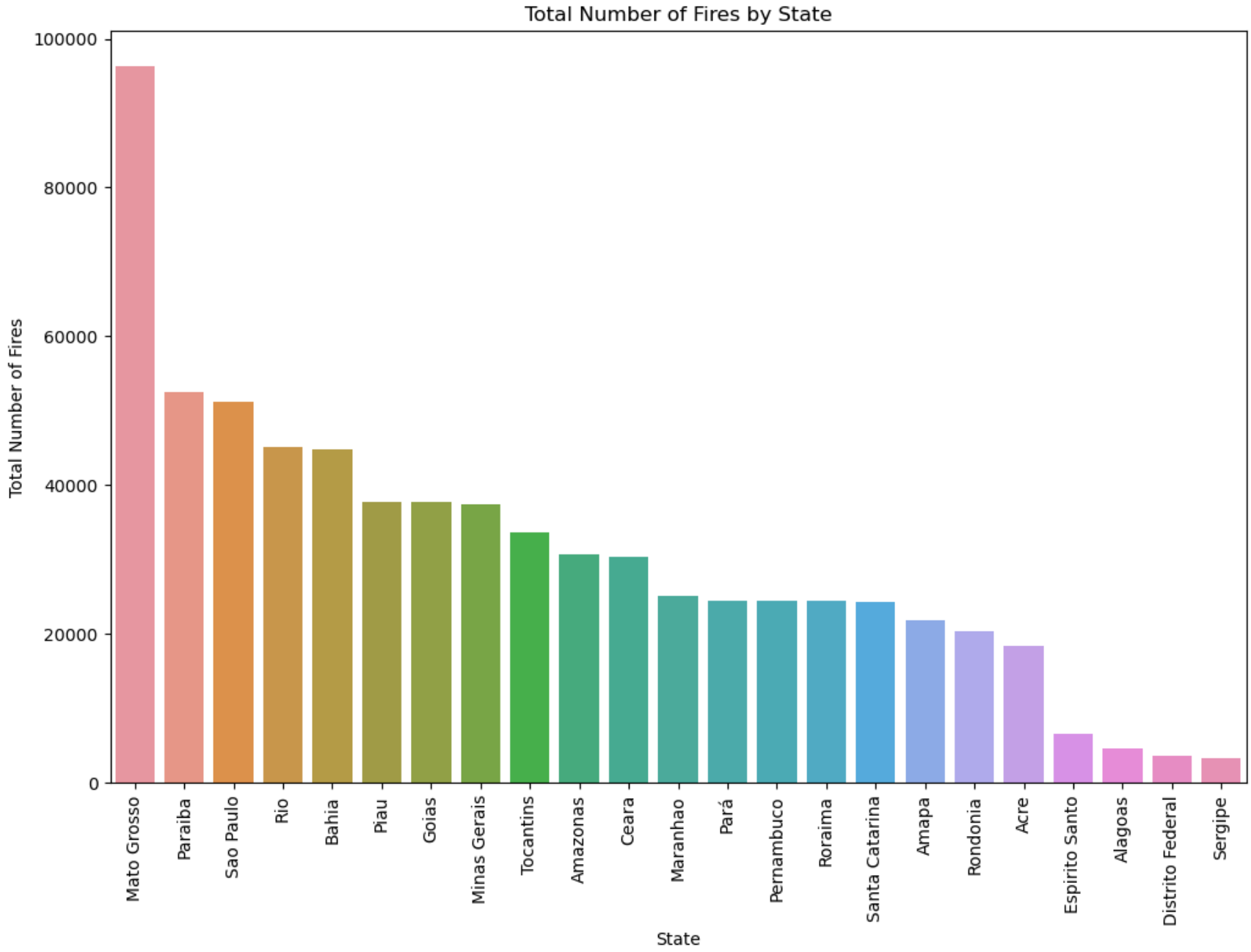
```
(array([15000., 20000., 25000., 30000., 35000., 40000., 45000.]),
 [Text(0, 15000.0, '15000'),
  Text(0, 20000.0, '20000'),
  Text(0, 25000.0, '25000'),
  Text(0, 30000.0, '30000'),
  Text(0, 35000.0, '35000'),
  Text(0, 40000.0, '40000'),
  Text(0, 45000.0, '45000')])
```



In [11]:

```
fps = pd.DataFrame(Forest_fires.groupby('state')['number'].sum().sort_values(ascending=False).reset_index())
fig,ax = plt.subplots(figsize=(12,8))
ax = sns.barplot(data=fps,x='state',y='number')
ax.set(xlabel='State',ylabel='Total Number of Fires',title='Total Number of Fires by State')
```

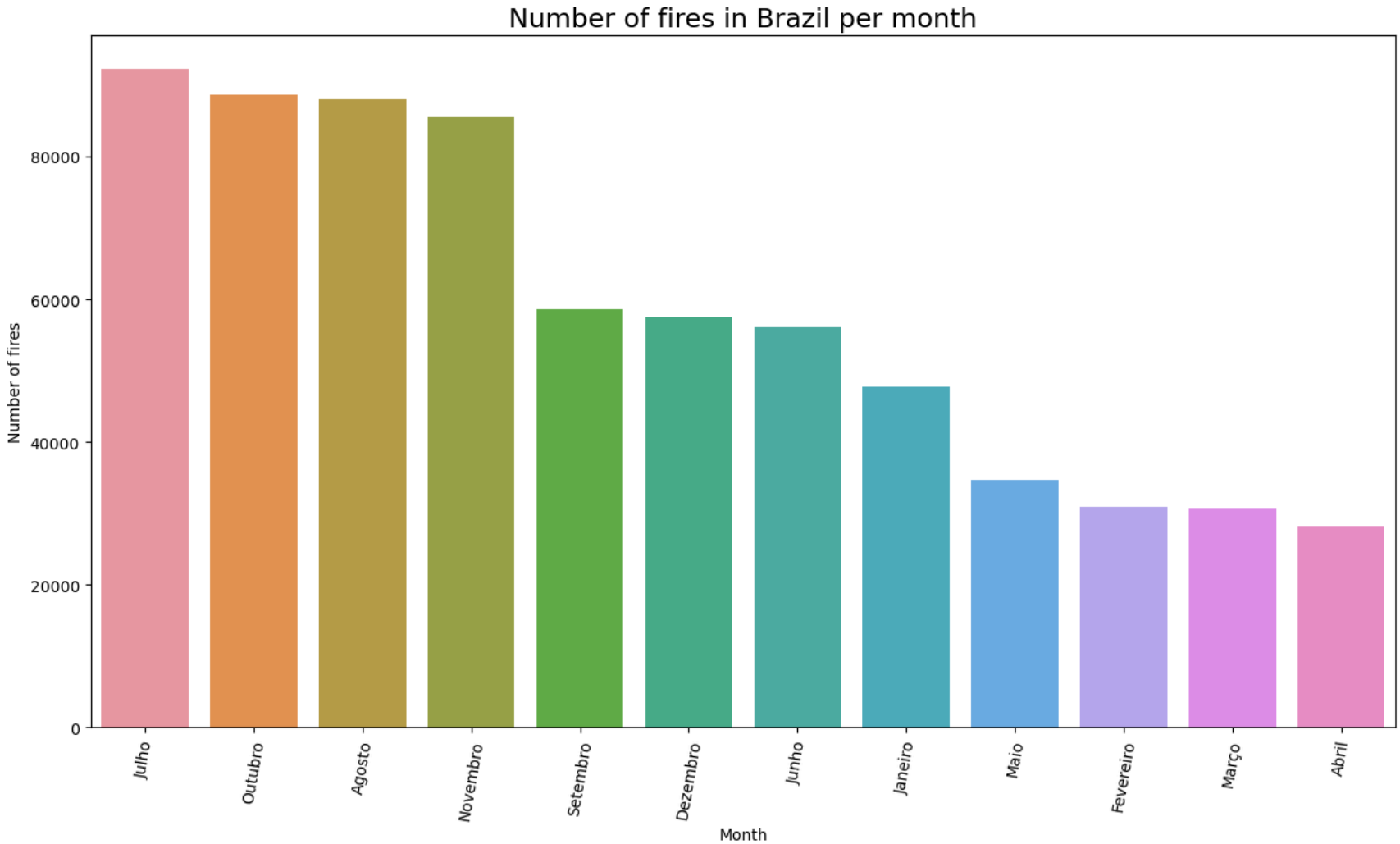
```
plt.xticks(rotation='vertical')
plt.show()
```



In [12]:
```
fpm = pd.DataFrame(Forest_fires.groupby('month')['number'].sum().sort_values(ascending=False).reset_index())

plt.figure(figsize=(15,8))
sns.barplot(data=fpm, x='month', y='number')
plt.title('Number of fires in Brazil per month', fontsize=17)
plt.xlabel('Month')
plt.ylabel('Number of fires')
plt.xticks(rotation=80)
```

Out[12]:
```
(array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11]),
 [Text(0, 0, 'Julho'),
  Text(1, 0, 'Outubro'),
  Text(2, 0, 'Agosto'),
  Text(3, 0, 'Novembro'),
  Text(4, 0, 'Setembro'),
  Text(5, 0, 'Dezembro'),
  Text(6, 0, 'Junho'),
  Text(7, 0, 'Janeiro'),
  Text(8, 0, 'Maio'),
  Text(9, 0, 'Fevereiro'),
  Text(10, 0, 'Março'),
  Text(11, 0, 'Abril')])
```



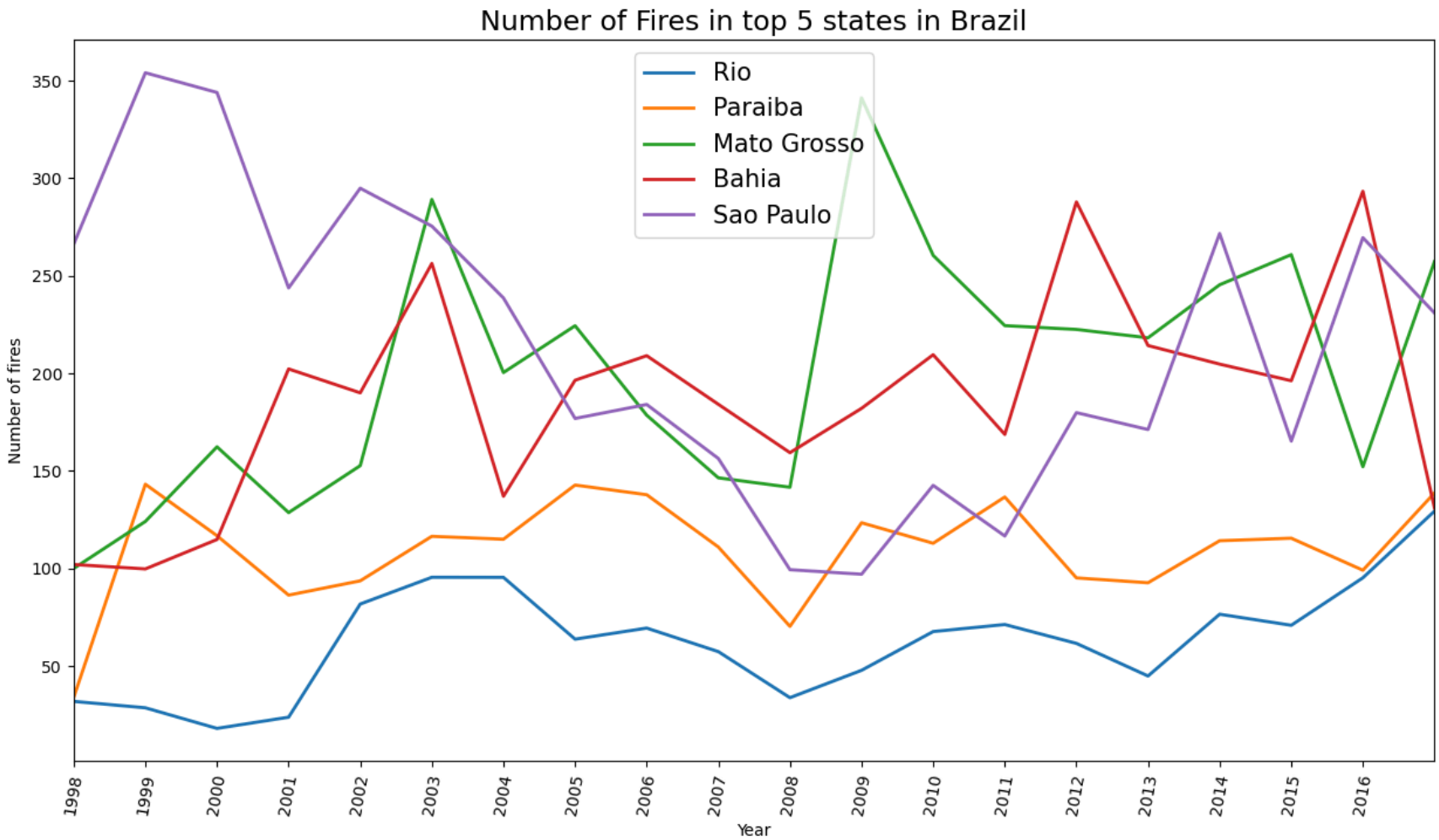
In [13]:
```
df1 = pd.DataFrame(data=Forest_fires[Forest_fires['state'] =='Rio'])
df2 = pd.DataFrame(data=Forest_fires[Forest_fires['state'] =='Paraiba'])
df3 = pd.DataFrame(data=Forest_fires[Forest_fires['state'] =='Mato Grosso'])
df4 = pd.DataFrame(data=Forest_fires[Forest_fires['state'] =='Bahia'])
df5 = pd.DataFrame(data=Forest_fires[Forest_fires['state'] =='Sao Paulo'])

plt.figure(figsize=(15,8))

df_list = [df1, df2, df3, df4,df5]
df_group = []
for x in df_list:
    x.groupby('year')['number'].sum().reset_index()
```

```
    df_group.append(x)

for x in df_group:
    sns.lineplot(x='year', y='number', data=x, label=x['state'].iloc[0],lw=2, errorbar=None)
    plt.title('Number of Fires in top 5 states in Brazil', fontsize=17)
    plt.xlabel('Year')
    plt.ylabel('Number of fires')
    plt.xticks(np.arange(1997,2017,1), rotation=80)
    plt.xlim(1998,2017)

plt.legend(fontsize=15)
```

Out[13]: <matplotlib.legend.Legend at 0x25dd7747e50>